Made by an AI persona. Aisha Rahman is one of SkillForge's AI personas, not a real person. AI wrote this project as an example for teachers.

Otter Dive: teacher guide

Pupils make a cardboard box with a crank. One cam inside lifts a creature slowly, then drops it. For Sec 1, 32 pupils in 16 pairs, in one 60-minute lesson. Any room with tables works. No code, no power.

Materials, per pair

- 1 frame strip, 375 × 90 mm box card, pre-cut and pre-scored
- 1 template on A4 card, printed at 100%
- 2 bamboo skewers, tips snipped flat
- 1 flat wooden coffee stirrer, 1 drinking straw
- Masking tape, 1 ruler, 2 pairs of scissors, markers

Shared by the class: 2 low-temperature glue guns on silicone mats, at one station. Also PVA, and a ball of Blu Tack per table to poke holes into.

Before the lesson, about 45 minutes

- Cut and score 16 frame strips with a craft knife on a cutting mat. Cut away from your hand.
- Cut the slot in each top panel to fit your stirrers.
- Snip the points off 32 skewers with side cutters.
- Build one demo. Turn it in front of the class.

Safety

- Pupils use scissors only. All knife work is done before class.
- Skewers stay below chin height. No sword fights: a skewer waved is a skewer taken.
- Hot glue only at the station, one pupil per gun, nozzle on the mat. Burns: cool running water for 20 minutes, then tell me.
- Poke holes with a pencil into Blu Tack, never towards a hand.

The 60 minutes

0 to 5	Hook. I turn the demo. "What made it drop so fast?" Pairs guess.
5 to 10	Safety rules and roles: one cutter, one builder. Swap at minute 30.
10 to 20	Fold the tube, tape the tab. Poke axle holes. Add straw bearings.
20 to 30	Pick a creature and a cam to match its story. Cut 2 cam layers, glue, thread the skewer.
30 to 40	Foot on the stirrer. Slide it through the slot. Fix the creature and wave.
40 to 48	Test: 3 full turns. Measure travel with a ruler. Fix one problem.
48 to 55	Swap boxes. Turn another pair's build and leave one sticky note.
55 to 60	Prompts 1 and 2 on the sheet. Skewers in the cup, glue guns off and unplugged.

Pitfalls and fixes

- **Cam slips on the axle.** Two layers of card, a tape flag each side.
- **Axle wobbles.** Hole too big. Add a card washer, or move the hole.
- **Creature spins.** Use the flat stirrer, not a skewer.
- **Snail cam jams.** It turns one way only. Draw a "this way" arrow.
- **Foot catches on the cam.** The cam rides off a small foot. Use the 30 mm disc.
- **Box squashes.** Tape the tab along its whole length.
- **Travel is less than the template says.** Card squashes. Ask why, then move on.

To make it easier, give out cams already cut. To make it harder, put two cams on one axle, half a turn apart.

Sources: burn first aid from KK Women's and Children's Hospital. Snail cams turn one way only: Rob Ives, "Cam design". Template and build record: the Otter Dive project page.